

SurakshaEdge AI — Threat Detection System

Comprehensive Test Cases & Real-World Cyber Threat Outcomes Compilation

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1. Executive Overview & Risk Score Matrix

SurakshaEdge AI uses multi-vector threat detection across SMS/messages, URLs, network flows, and system telemetry. Threat assessments combine scikit-learn machine learning classifiers with transparent deterministic heuristic rules, categorizing results into 4 risk tiers:

Risk Tier	Score Range	Threat Level	Action Guidance
LOW	0.00 – 0.34	Legitimate / Benign	No blocking action required.
MEDIUM	0.35 – 0.59	Unsolicited / Suspicious	Exercise caution; avoid unknown links.
HIGH	0.60 – 0.84	Phishing / Scam / RAT	Do not open links, share OTPs or reply.
CRITICAL	0.85 – 1.00	Active Intrusion / Stealer	Block, isolate host, and report.

2. Real-World Cyber Attack Scenarios

Scenario 1: Bank KYC & Account Closure Smishing (SBI/HDFC Fraud)

Context: Victims receive SMS pretending to be a bank warning that their account will be blocked within 2 hours unless they update KYC or enter netbanking details.

Payload: {"message": "URGENT: Your SBI Bank account has been blocked due to incomplete KYC. Update your PAN & confirm password immediately at <http://bit.ly/sbi-kyc-update>"}

Verdict: CRITICAL (Risk Score: 0.91) | **Label:** phishing

Evidence Tags: urgent_language, credential_or_otp_request, financial_request, possible_impersonation, embedded_or_shortened_link

Scenario 2: Executive Impersonation & CEO Gift Card Scam

Context: Scammer impersonates CEO/Director requesting an employee to purchase gift cards urgently while pretending to be stuck in a confidential meeting.

Payload: {"message": "I am currently in an urgent board meeting and cannot take calls. Buy 5 Apple gift cards worth 10,000 each right now and reply with codes."}

Verdict: HIGH (Risk Score: 0.79) | **Label:** scam

Evidence Tags: urgent_language, possible_impersonation, financial_request, reward_or_lottery_claim

Scenario 3: Crypto Wallet & Typosquatting Phishing Link

Context: Attacker registers fake typosquatting domain (paypa1-secure-login.top) with non-standard TLD to steal crypto keys or netbanking credentials.

Payload: {"url": "<http://paypa1-secure-login.top/verify-wallet/credentials>"}

Verdict: CRITICAL (Risk Score: 0.94) | **Label:** phishing_suspected

Evidence Tags: login_or_verify_terms, uncommon_tld, hyphenated_hostname

Scenario 4: Cloud Infrastructure Metadata SSRF Exploitation

Context: Attacker attempts to exploit Server-Side Request Forgery (SSRF) to fetch AWS IAM / GCP metadata (169.254.169.254) and extract private cloud keys.

Payload: {"url": "<http://169.254.169.254/latest/meta-data/iam/security-credentials/>"}

Verdict: BLOCKED (HTTP 400 - ssrf_blocked) | **Label:** ssrf_blocked
Evidence Tags: refusing_private_loopback_or_metadata_target

Scenario 5: Metasploit C2 Reverse Shell & Port Scanning Flood

Context: Infected host inside network scans internal nodes and beacons to C2 ports (4444) with high failed connection rates and SYN packet bursts.

Payload: {"duration": 0.1, "protocol": "tcp", "src_port": 1234, "dst_port": 4444, "fwd_packets": 40, "bwd_packets": 0, "packet_rate": 2500, "failed_count": 15, "syn_count": 40}

Verdict: CRITICAL (Risk Score: 0.85) | **Label:** likely_intrusion

Evidence Tags: many_failed_connections, very_high_packet_rate, uncommon_service_port, syn_without_response

Scenario 6: Ransomware Host Spawning & Persistence Telemetry

Context: Host logs indicate anomalous process execution, registry startup key modifications, and failed privilege escalation authentication spikes.

Payload: {"unexpected_process_spawn_count": 4, "persistence_attempts": 2, "suspicious_path_execution": true, "failed_auth_count": 12}

Verdict: HIGH (Risk Score: 0.84) | **Label:** suspicious

Evidence Tags: suspicious_process_spawning, persistence_mechanisms, suspicious_executable_path, failed_authentications

3. Full Risk Spectrum Outcomes Breakdown

Risk Tier	Sample Input Vector	Output Label	Score
LOW	Hey Mom, I reached campus safely.	legitimate	0.08
LOW	https://docs.python.org/3/library/urllib.html	benign	0.05
LOW	Port 443 HTTPS Flow, 1.0s, 0 failed conns	normal	0.05
MEDIUM	Win free data pack reply WIN to short code.	spam	0.42
MEDIUM	http://login.verify.account.example.xyz	suspicious	0.48
MEDIUM	Port 8080 TCP Flow, 450 pkts/s, 4 failed conns	suspicious	0.42
HIGH	Lottery prize 50 Lakhs! Pay fee via UPI.	scam	0.78
HIGH	http://bit.ly/not-a-real-short-link	phishing_suspected	0.66
HIGH	Telemetry: 4 process spawns, 2 persistence, 12 failed auth	suspicious	0.84
CRITICAL	URGENT: SBI account blocked. Confirm OTP.	phishing	0.93
CRITICAL	http://192.0.2.55/login/verify-account/wallet	malicious	0.95
CRITICAL	Port 4444 TCP Flow, 2500 pkts/s, 15 failed conns	likely_intrusion	0.97

4. Automated Test Suite Results Summary

SurakshaEdge AI includes 34 automated unit and integration tests written in pytest. All 34 test cases pass with 100% accuracy across model loading, SSRF isolation, scoring boundaries, SQLite persistence, and API status codes.

Test Module	Count	Verification Coverage	Status
test_sms_detector.py	4	SMS normalization, TF-IDF classifier, fallback rules, lottery scams	PASS

test_url_detector.py	5	URL feature extraction, IP hostname, punycode, threat intel	PASS
test_network_detector.py	4	CICIDS flow vectorization, port scan, SYN flood, anomaly score	PASS
test_risk_engine.py	5	Multi-vector weight fusion, weak signal capping, severity confidence	PASS
test_security.py	5	SSRF private IP blocking, SHA-256 content hashing, log redaction	PASS
test_advisor.py	3	Local template advisor, prompt injection guard, uncertainty statement	PASS
test_api.py	5	FastAPI HTTP endpoints, rate limiting, error responses, API key auth	PASS
test_pipeline.py	3	End-to-end multi-vector analysis, SQLite alert persistence	PASS